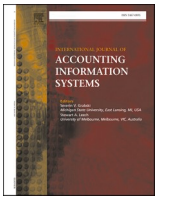
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Enhancing continuous auditing with large language models: AI-assisted real-time accounting information cross-verification☆

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ABSTRACT

This study proposes a Large Language Model (LLM)-assisted framework for parsing near real-time audit evidence from text to cross-verify accounting data. This addresses a core challenge in continuous auditing (CA): analyzing textual data in near real-time. The three-step framework involves 1) Preprocessing text, 2) LLM inference guided by auditor objectives/schemas and prompts, and 3) Validating accounting records against LLM-derived audit evidence. Demonstrated on a real-life Brazilian governmental payroll system, the framework cross-verifies payroll data with human resources information from the Official Gazette. Compared to auditors' existing process, the framework improves audit effectiveness through full population testing with cost efficiency, decreases the cross-verification time by up to 83%, and achieves a high extraction accuracy of 97%. The study demonstrates the advantage of LLMs in facilitating audit quality and contributes by proposing an LLM-assisted framework to facilitate practical CA application. Beyond showcasing the potential of LLMs, this study contributes by structuring their application into a replicable three-step framework that auditors without advanced technical expertise can adopt. By aligning LLM capabilities with auditor-defined objectives, parsing schemas, and rule-based cross-verification, the framework ensures that efficiency gains translate into effectiveness, replicability, and audit quality in real CA environments.

1. Introduction

In the rapidly evolving landscape of auditing, the emergence of continuous auditing (CA) (Vasarhelyi and Halper, 1991) represents a significant stride toward more dynamic and reliable oversight. CA is the generation of written assurance simultaneously or a short period of time after the events to be assured (American Institute of Certified Public Accountants (AICPA), 1999). It is a method used to perform control and risk assessments automatically on a more frequent basis (Shilts, 2017). Prior studies have extensively discussed

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the various benefits of CA adoption, including improvements in internal controls and more timely response to errors and anomalies (Kuenkaikaew, 2013; Warren Jr et al., 2015). Against this backdrop, this study develops and demonstrates a large language model (LLM)-assisted framework that enables auditors to draw audit evidence from textual sources and cross-verify it against accounting records in near real-time, extending CA to textual data it has not been able to monitor continuously.

Despite CA's perceived advantages, its adoption in practice has been slower than anticipated. According to a recent survey, 70% of internal audit function (IAF) professionals in Germany, Austria, and Switzerland reported not using CA, and 50% of them indicated no plans to adopt it (German Institute of Internal Auditors, 2023). One of the most persistent impediments to the adoption of CA in practice is the need for dependable and uninterrupted access to data (Earley, 2015; Hardy and Laslett, 2015; Rikhardsson and Dull, 2016). Management often perceives CA as an additional organizational burden, both because its benefits are difficult to quantify and typically materialize only over an extended period of time and because the cost and effort required to conduct CA can themselves be substantial (Eulerich et al., 2023; Eulerich et al., 2025). In addition, auditors are often reluctant to rely on information generated by automated systems, including artificial intelligence (AI) systems, due to prevailing negative perceptions toward disruptive technologies (Commerford et al., 2022; Commerford et al., 2024; Kokina et al., 2025).

Beyond these organizational barriers, a more fundamental constraint lies in the type of data CA has relied on. CA systems have traditionally used structured data as inputs, particularly the transactional and master data housed within enterprise resource planning (ERP) systems, because CA requires near real-time or frequent audits by utilizing data modeling and analytics for monitoring (Chan and Vasarhelyi, 2011). As a result, most CA applications and literature have been built on transactional data and system logs extracted from ERP systems (Alles et al., 2008; Singh et al., 2014). However, unstructured data such as text comprises most of the information within organizations, with some estimates suggesting it may account for up to 80% of all organizational data (Shilakes and Tylman, 1998; Gandomi and Haider, 2015). Limiting CA inputs to structured data may constrain the effectiveness of CA, as it overlooks insights embedded in unstructured resources, such as emails, contracts, policies, news articles, social media posts, announcements, and various disclosures. Prior studies have explored the advantages of incorporating textual data and textual analysis in accounting and auditing (Zhaokai and Moffitt, 2019; Alzamil et al., 2020; Bochkay et al., 2023). Incorporating textual data as input can significantly expand the scope of CA and its benefits. Because CA's benefits remain difficult to quantify while its costs are substantial, broadening scope delivers real value only when those costs also fall. By simplifying and largely automating the process, the framework this study develops reduces these costs, so that the wider benefits of drawing on textual data become worth pursuing.

However, the vast volume of textual data, together with the limitations of existing text-mining approaches, makes near real-time qualitative analysis difficult for auditors (Duan et al., 2022; Geerts and O'Leary, 2022). The key obstacle is not the file format of textual data, since digitally generated documents are often readily made machine-readable. Rather, the key difficulty lies in the content itself, where the same fact appears in highly variable forms, alongside typographical errors, domain-specific legal jargon, and details scattered across multiple sentences. Existing text-mining techniques compound the difficulty, having been designed for high-level, aggregative analysis such as sentiment analysis and topic modeling (Loughran and McDonald, 2016; Tetlock, 2007), while rule-based methods such as regular expressions, dictionaries, and named entity recognition break down at the slightest change in phrasing (Huq and Carling, 2024; Li et al., 2026). As a result, auditors cannot reliably parse granular audit evidence from text at scale and remain confined to a manual, sample-based review, forgoing the full population coverage that motivates CA in the first place.

To address this gap, this study proposes an LLM-assisted framework, developed as a design science research (DSR) artifact, that enables auditors to parse and collect audit evidence from text and cross-verify it against accounting records within a CA environment in near real-time. LLMs are well suited to this task, interpreting and extracting information from heterogeneous text with a contextual understanding that prior techniques could not achieve (Gu et al., 2024; Huang et al., 2023). They can be instructed in natural language through prompt engineering, which sharply lowers the technical barrier for auditors (Li et al., 2025), and can process multilingual documents (Kholodna et al., 2024). LLMs are prone to hallucination, however, generating responses that are not factually correct (Gu et al., 2024; Li et al., 2025). The framework is therefore designed so that auditors retain meaningful control over the evidence collection process through auditor-defined objectives, parsing schemas, and rule-based verification, rather than delegating assurance wholesale to LLMs. In line with DSR, we use the term framework to denote a conceptual structure that organizes problem identification, solution design, demonstration, and evaluation (Hevner et al., 2004; Geerts, 2011).

To demonstrate how the framework can be applied in auditing, we apply it to a real-life Brazilian state government payroll CA system. In this setting, the framework facilitates a near real-time cross-verification process, enabling internal auditors to compare payroll records (sourced from the personnel management system) with publicly accessible human resources-related information in the Official Gazette (DOE-*Diário Oficial do Estado*).[1] DOEs contain information in Portuguese on various topics within public administration, including data pertaining to public servants. The objective of cross-verification is to identify risks such as the existence of "ghost employees" who are on the payroll but have never been officially appointed in DOE. Section IV details the full set of cross-verification risks. Identifying these risks serves the internal auditors' immediate objective of confirming that each payroll record is supported by and consistent with the personnel acts published in the DOE. For the state government, that confirmation serves the broader goal of avoiding the financial and reputational consequences of undetected inconsistencies. Even a few instances might suggest a lack of control and negatively impact the public's perception of the government (Barros, 2025).

[1] In Brazil, the Official Gazette (DOE) is a channel where laws, bids (governmental purchasing information), personnel acts (information regarding governmental human resources), and all other activities of a Brazilian government administrative division are published. DOE is the official publication of various acts of the government, including federal, state, and municipal, and serves as an official document for almost all Brazilian public organizations. Some examples of DOE can be found here: State of Santa Catarina (https://portal.doe.sea.sc.gov.br/v2.40.00/#/portal/edicao/buscar-edicao), City of Florianópolis (https://www.pmf.sc.gov.br/governo/index.php?pagina=govdiariooficial), Federal Government (https://www.in.gov.br/inicio).

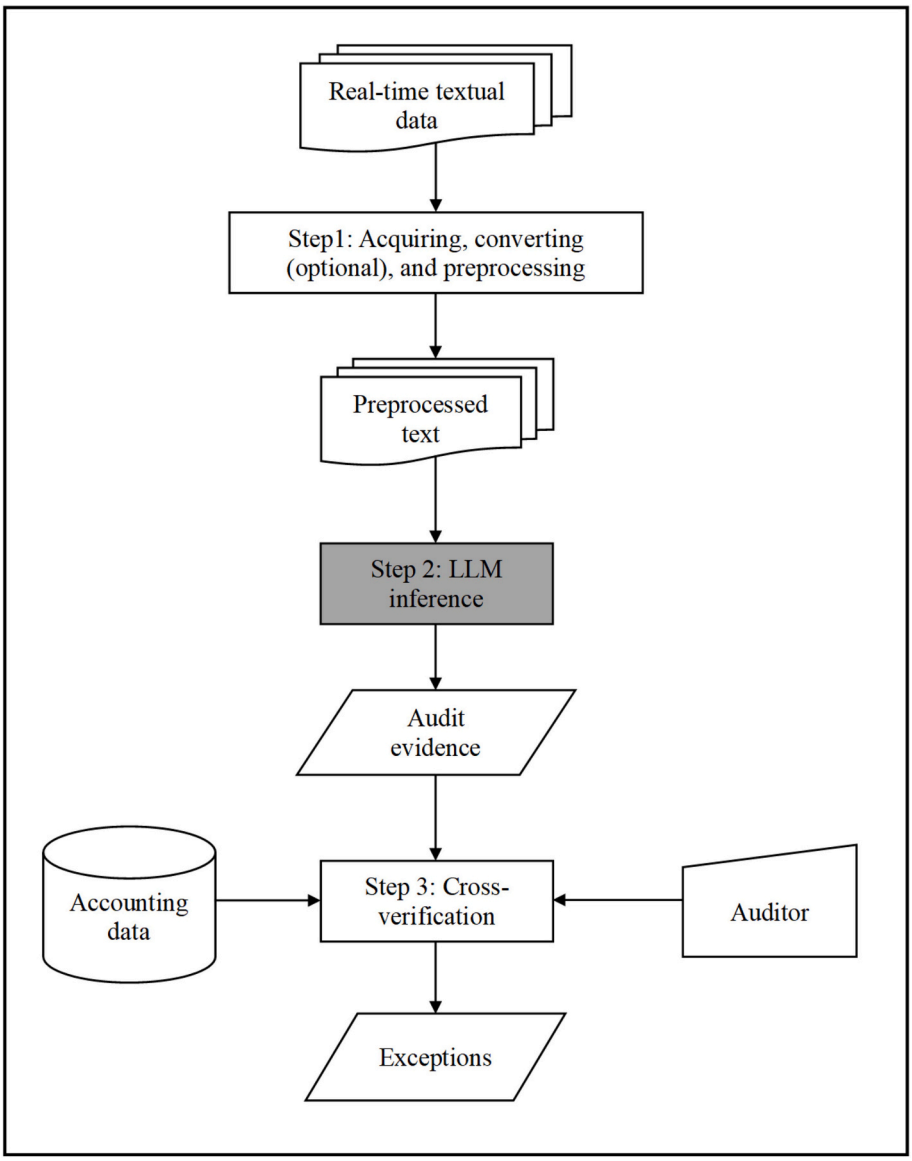


Fig. 1. Flowchart of the Framework.

The framework operates in three steps (Fig. 1). It first acquires, converts, and preprocesses the textual source, such as the daily DOE, using automated scraping and robotic process automation (RPA) (Step 1). It then applies an LLM to extract critical audit evidence, for example an appointment, a position, or a penalty, guided by auditor-defined objectives and task-specific parsing schemas (Step 2). Finally, auditors apply a rule-based cross-verification that reconciles the accounting records against the parsed evidence and flags exceptions (Step 3). This replaces a manual workflow in which auditors must locate the relevant DOE editions, read or keyword-search lengthy PDFs, and validate each finding by hand, which time constraints render infeasible at full population coverage.

The demonstration is evaluated in three dimensions, namely effectiveness, efficiency, and quality of cross-verification. The results indicate that the framework can significantly enhance auditors' CA effectiveness by switching from random sampling examination to full population testing (No, 2019) without incurring significant costs, thereby enhancing the detection of low-frequency, high-risk errors (Freiman et al., 2022) and mitigating reputational risks. Overall, the framework enhances efficiency and quality by reducing the average cross-verification time for a single piece of accounting information by up to 83% compared to the existing process and achieving an average accuracy of 97% for the parsed audit evidence.

This study contributes to accounting literature and practice. It is the first study proposing an LLM-assisted framework to enhance the CA process. With the near real-time generation of text across all business activities and systems, existing practices face difficulties in accurately and efficiently analyzing them and parsing audit evidence. This study contributes by proposing a novel framework that utilizes LLMs to collect near real-time audit evidence from text, an essential step to applying CA in practice. It overcomes the

limitations of prior text-mining methods, which can only perform aggregate analysis (e.g., sentiment analysis and topic modeling), are sensitive to variations in expression (e.g., Python Regex), or are limited in item scope (e.g., dictionaries and named entity recognition). The framework enhances audit quality through full population cross-verification and facilitates the practical application of CA. Researchers designing advanced CA models and applications would be interested in applying this framework as one critical step to obtain near real-time textual audit evidence.

This study also contributes to practice by providing stakeholders with an effective process. Although it is demonstrated only on one governmental payroll system, the application of this framework extends far beyond this specific system. Large and mid-size entities would benefit from applying this framework within IAFs to reduce fraud and operational costs. External auditors could also leverage this framework to obtain additional audit evidence from various textual sources, thus enhancing audit quality. This study is also relevant to audit regulators who might recognize the advantage of generative AI technologies in enhancing audit quality and providing guidance on its adoption.

Finally, while LLMs provide the core technology, using them without a well-designed structure can lead to inconsistent results and misalignment with audit objectives. Auditors typically lack the technical background to independently design prompts, preprocessing routines, or cross-verification steps, limiting practical adoption. Our structured three-step framework addresses this gap by operationalizing LLM capabilities into a systematic process including preprocessing textual evidence, guiding inference through auditor-defined schemas and prompts, and validating results via rule-based cross-verification. This design ensures replicability, quality assurance, and alignment with domain-specific assurance tasks, extending beyond efficiency to strengthen audit effectiveness, transparency, and trustworthiness.

The remainder of this paper is organized as follows. Section II introduces the background and reviews relevant prior studies. Section III proposes the framework. The demonstration of the framework is presented in Section IV, and Section V evaluates the results. Section VI discusses and concludes the paper.

2. Large language models for continuous auditing

CA, first proposed by Groomer and Murthy (1989) and Vasarhelyi and Halper (1991), applies information technology to standard audit processes, enabling auditors to shift from retrospective testing to a near real-time approach (Alles et al., 2002; Eulerich and Kalinichenko, 2018). Prior research has implemented a range of technologies to support CA. For example, Yoon et al. (2021) introduced a three-layer CA system that aggregates journal entry-level transactions to identify control deviations, and Zhang et al. (2022) presented a continuous control monitoring framework combining interactive visualization and machine learning to reduce information overload.

A key requirement for CA is the real-time or frequent analysis of data (Chan and Vasarhelyi, 2011), and recent work has incorporated alternative real-time sources into continuous monitoring, including social media sentiment (Duan et al., 2023), Internet-of-Things and big data for environmental, social, and governance assurance (Gu et al., 2023), and weather data to corroborate financial information (Yoon et al., 2024).

However, dealing with a large volume of real-time textual data is still challenging (Geerts and O'Leary, 2022). Typical examples involve analyzing system logs, procurement agreements, news releases, and daily reports. Although this data contains crucial evidence for audit purposes (Appelbaum et al., 2017; Yoon et al., 2015), auditors are limited by their ability to manually examine it. Existing natural language processing techniques, such as regular expressions (Li et al., 2026), bag-of-words (Purda and Skillicorn, 2015), topic modeling (Senave et al., 2023), and sentiment analysis (Alzamil et al., 2020), have limited ability to parse granular information from human language that usually varies in expressions. In addition, language barriers can further complicate auditors' evidence collection from foreign clients (Burke et al., 2020).

LLM usage has emerged as a technique to facilitate real-time textual data processing in the CA process. LLMs operate through natural language and possess basic reasoning abilities in multiple languages, substantially enhancing the effectiveness and efficiency of textual analysis in accounting research (de Kok, 2025; O'Leary, 2023a, 2023b, 2024). They offer significant benefits to accounting by efficiently addressing various textual analysis tasks, including extracting information (Huang et al., 2023; Li et al., 2026), identifying risks in financial statements (Gu et al., 2023; Kim et al., 2023), and generating or refining audit reports (Emett et al., 2025; Eulerich and Wood, 2025).

Although research lacks a direct application of LLMs to the CA process, several recent studies point the way. Gu et al. (2024) introduce the concept of AI co-piloted auditing, highlighting the potential of foundation models to reform audit procedures. Li and Vasarhelyi (2024) compare the mainstream LLM implementation methods in accounting, user interface (UI) and application programming interface (API), and introduce their integration with RPA, offering insights into the advantages, limitations, and costs of each. This study builds on their API-RPA approach. While these studies highlight the potential of LLMs for accounting tasks, their use remains largely unstructured and exploratory. This study advances the literature by translating LLM capabilities into a structured, auditor-oriented framework, designed as a DSR artifact, that bridges the gap between experimental applications and replicable adoption in real continuous auditing settings.

3. Research design and framework

This study follows the DSR methodology (DSRM) to develop and test the framework (David et al., 1999; Geerts, 2011; Gregor and Hevner, 2013; Hevner et al., 2004; Peffers et al., 2007; Van Aken, 2005). Unlike research that seeks to explain existing phenomena, DSR focuses on creating artifacts that solve practical problems, an orientation that has been advocated for accounting research

intended to inform practice (Geerts, 2011). The framework is first developed in this section and subsequently demonstrated with a real-life application in the following section.

3.1. Objectives of the framework

The study aims to enhance CA by creating an LLM-based framework for real-time cross-verification of accounting data. Specifically, it leverages LLMs' language comprehension ability to parse critical audit evidence from textual data to cross-verify accounting records. Existing audit techniques face difficulties in parsing audit evidence from real-time textual data, which usually results in low efficiency and reduced audit quality. Following prior research that evaluates the use of innovative technology to enhance the audit process (Christ et al., 2021), this study aims to improve the effectiveness, efficiency, and quality of cross-verification.

To measure effectiveness, this study evaluates the scope and validity of the cross-verification compared to auditors' existing workflows and whether the results contribute to their identification of irregularities in the CA process. To measure efficiency, the study assesses the time saved in cross-verifying the same amount of data using the framework compared to auditors' existing approach. Finally, to measure quality, the study evaluates the accuracy of the parsed audit evidence by the framework against the original textual data.

3.2. Design of the framework

3.2.1. Acquiring, converting, and preprocessing (Step 1)

This study designs and develops a three-step framework to enhance the CA process with LLMs (see Fig. 1). *Step 1* involves first acquiring and converting the real-time textual data into machine-readable text, such as plain text (TXT) files. Continuously generated files from an external department, website, or system are beneficial for automatic acquisition using a programming approach. Data that is continuously posted on websites, such as social media or organizational portals, can be scraped programmatically. If the data is generated by a local system that does not support programming-level operations, an RPA bot can be built to automatically crop or download relevant textual data. Prior studies have introduced various programming and robotic approaches to automate data-acquiring tasks from textual sources (Debreceny et al., 2019; Li and Vasarhelyi, 2024; Wang, 2022; Zhang, 2023).

Data conversion is not necessary for all the textual data, as some can be read directly. For instance, the news releases of companies, such as the text in Rich Text Format, are usually machine-readable and can be directly processed by LLM. However, some reports, such as those issued in scanned documents or handwritten formats, usually require data conversion before they can be imported and further analyzed by the LLMs (Duan et al., 2022; El-Haj et al., 2020; Li and Vasarhelyi, 2024).[2]

Finally, data preprocessing should be implemented to eliminate inconsistencies in the textual data. For instance, lowercase can be enforced on all text to facilitate keyword matching. Redundant spaces, punctuation, or special characters can be eliminated to enhance the subsequent LLM inference accuracy and consistency.[3] Senave et al. (2023) and Bochkay et al. (2023) provide a comprehensive review of text preprocessing steps in accounting research.

3.2.2. LLM inference (Step 2)

The LLM Inference step is the most important one, as it involves implementing LLMs as the key enabling technology for parsing audit evidence for cross-verification (see *Step 2* in Fig. 1). As such, an in-depth illustration is provided in Fig. 2. Specifically, *Step 2* applies LLMs to parse critical information from the preprocessed text to form audit evidence. With the audit tasks, the auditors should define a set of schemas for parsing the relevant information from the text (*Establishing parsing schema* in Fig. 2). The schemas are audit task-oriented and are usually relevant for auditors to obtain critical evidence for substantive analytical procedures. The adoption of schema-guided extraction, as opposed to alternatives such as retrieval-augmented generation that queries unstructured text on demand, or direct record-to-text comparisons without an intermediate schema, is motivated by the requirements of the CA context. First, mapping textual evidence into predefined, auditor-validated fields creates a transparent audit trail, meeting regulatory documentation requirements for evidence provenance. Second, the structured output integrates directly with existing rule-based CA systems (e.g., the structured query language-based cross-verification engines used by the state government), whereas unstructured approaches would require LLM invocation at every comparison step. Third, in a CA environment where documents are generated daily, extracting structured fields once allows the parsed evidence to serve multiple audit objectives (e.g., ghost employee detection, pay-grade verification, leave compliance) without repeated LLM calls, substantially reducing computational costs and latency. Fourth, requiring auditors to define the schema in advance constrains LLM outputs to pre-validated fields, mitigating hallucination risks and ensuring alignment with audit objectives.[4]

For example, the employee appointment date, penalty amount, and paid leave time disclosed in DOE are parsing schemas for payroll audit. Once the schema is determined, auditors can generate a list of keywords for each. These keywords are used to filter the

[2] Conversion methods such as optical character recognition for scanned documents or handwriting recognition for handwritten files can be employed to make these reports machine-readable and suitable for processing by LLMs.

[3] Example Python codes on data preprocessing can be found extensively in academic research and online posts (Davydova, 2018; SRK, 2019; Wang, 2022).

[4] We acknowledge that this approach involves trade-offs: the predefined schema may not capture unanticipated audit-relevant information, and schemas must be updated as objectives evolve. However, in highly formalized settings where personnel actions and their audit implications are well understood, these trade-offs are outweighed by the benefits of structured, reusable, and auditable evidence extraction.

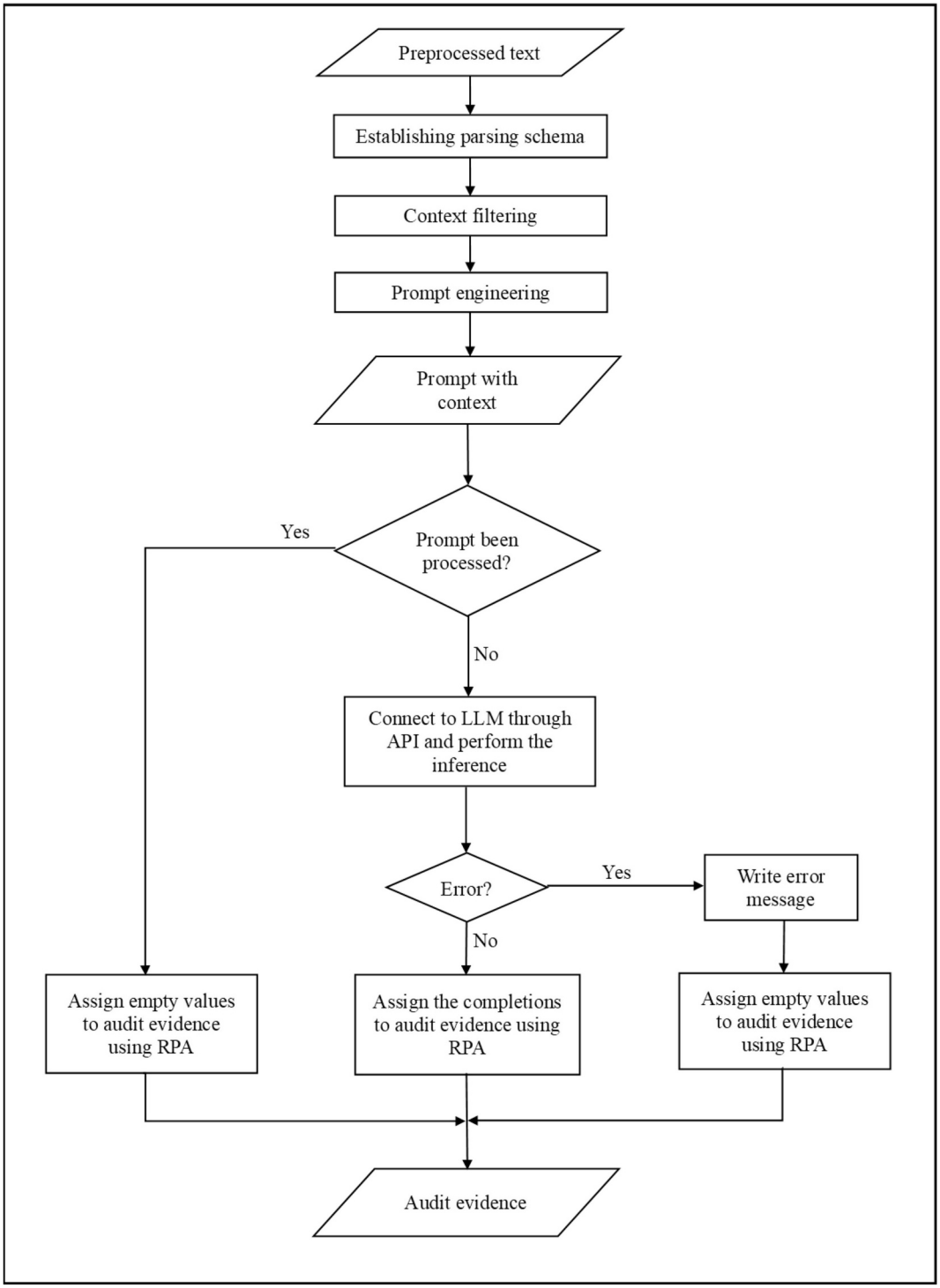


Fig. 2. Flowchart of Step 2: LLM Inference.

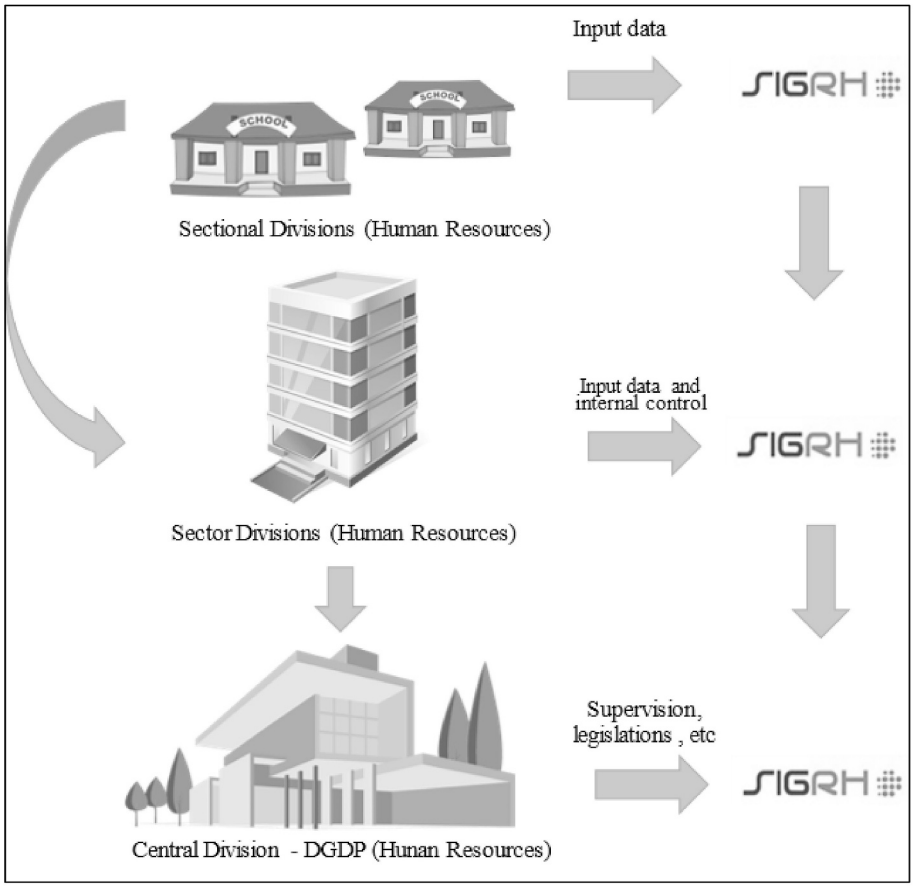


Fig. 3. Data Flow in the Payroll System of the State of Santa Catarina.

entire text to obtain the relevant paragraphs or sentences to each schema (*Context filtering* in Fig. 2). This avoids entering too much irrelevant information into the LLM, thus increasing the parsing accuracy and consistency, as well as decreasing the computational cost.[5]

After obtaining related paragraphs/sentences (*context*) for audit evidence, auditors should design the question (*prompt*) to be entered into the LLMs. This ensures the response (*completion*) is received in the desired direction and format (*Prompt engineering* in Fig. 2). Rather than fine-tuning LLMs, which requires extensive technical skills and hardware, prompts can be designed systematically to enhance the accuracy and consistency of completions. This study applies instruction tuning, which structures prompts into explicit sections specifying the role, background information, parsing methods, schemas, and output formats (Chung et al., 2024; Li et al., 2026). It also uses zero/one/few-shot learning, which supplies none, one, or more examples when the information to be parsed follows a specific format that examples can represent (Logan IV et al., 2021; Zhao et al., 2021).

The prompt engineering process usually involves several rounds of modifications based on evaluating the completions received from the LLMs. The best practice for auditors is to start with a simple prompt and examine the completions based on the audit tasks and objectives. If the completion identifies irrelevant information from the text, auditors should provide more detailed instructions in the prompts and may need to supply several examples. If the completion's output format is not consistent, a JavaScript Object Notation (JSON) schema and several output examples can be supplied to maintain the output format. An illustration in Section IV offers further insights for auditors to design and engineer the prompts.

When the prompts are prepared, auditors can input them with context into LLMs for completion, a process known as "inference" for the LLMs. Li and Vasarhelyi (2024) compared the two mainstream implementation approaches, UI and API, and proposed integrating each with RPA (*UI-RPA* and *API-RPA*) to overcome their limitations in scaling and in connecting to internal systems. Given that CA typically involves integrating multiple operational processes and transforming data across various applications, this study applies the *API-RPA* approach to perform the inferences. The detailed *API-RPA* inference process, including the RPA bot setup, timer-based

[5] Although LLMs can directly read long texts, the accuracy of their output may be compromised when the input text is excessively lengthy (Li et al. 2025). Additionally, Li and Vasarhelyi (2024) show that the context length is correlated with the cost of using LLMs, leading to higher computational expenses as the length of the input text increases.

scheduling, and small-group testing before large-scale deployment, is provided in Online Appendix A.[6]

A continuous process will encounter certain errors or exceptions during implementation. Therefore, the framework includes relevant pre-checking and error-handling functions that can help avoid processing repetitive tasks and prevent the process from failing when errors occur (*Prompt been processed* and *Error* checks in Fig. 2).

3.2.3. Cross-verification (Step 3)

In *Step 3* of Fig. 1, auditors can use the parsed audit evidence to check the compliance or inconsistency of the accounting data. For instance, the employee ID and names can be matched from two sources using a rule-based system to compare the documented pay grade and actual payment amount. Any exceptions discovered by the cross-verification processes can be further examined by the auditor. Feedback from the auditors could also serve as the basis for recursively optimizing the framework, such as adding more schemas or removing less risky ones. If too many exceptions are encountered, exception ranking based on suspicious scores (Issa, 2013), visualization (Zhang et al., 2022), or data aggregation (Yoon et al., 2021) could be applied to eliminate or prioritize these exceptions.

4. Illustration of the framework

The framework is tested to audit the payroll data of the Executive Branch of the State of Santa Catarina (PEESC-*Poder Executivo do Estado de Santa Catarina*), a Brazilian state government. More specifically, the daily generated DOE of Santa Catarina is used as the textual source to cross-verify the accounting records from the state's Human Resources Management System (SIGRH-*Sistema Integrado de Gestão de Recursos Humanos*), thus identifying potential risks. These risks include 1) the existence of "ghost employees" who are on the payroll but have never been officially appointed in DOE; 2) payment of funds in amounts incompatible with the function/position documented in DOE; 3) public servants penalized with loss of salary (or similar situations), but who are still receiving remuneration; 4) public servants on leave without entitlement to remuneration, but who are still receiving remuneration. Beyond the financial risk, the existence of significant inconsistencies also represents a substantial reputational risk to the state government.

4.1. Background of the State of Santa Catarina human resources and payroll systems

The Executive Branch of the State of Santa Catarina has a hierarchical administrative structure for managing human resources and payroll across all state departments, as shown in Fig. 3. The State Department of Administration oversees the Directorate of Human Resources Management and Development (DGDP-*Diretoria de Gestão e Desenvolvimento de Pessoas*). The DGDP regulates policies and maintains the SIGRH system, which houses almost all human resource data. As of November 2023, this complex system manages payroll for over 160,000 employees and 15,000 pensioners, disbursing over $500 million per month (Transparency Portal, 2023).[7] Payroll is governed by an extensive legal framework stipulating payment regulation for each type of employment or payment category.

The Personnel Audit Management (GAPES-*Gerência de Auditoria de Pessoal*) function is the internal audit team for the state payroll system, tasked with auditing various aspects related to the functional life of public servants within the PEESC. GAPES has more than 10 years of experience using CA techniques for payroll, with recent efforts focused on modernizing and automating the process. The current approach prioritizes rules-based auditing, employing automated structured query language scripts executed monthly. Although the model is well-established, it is confined to data from SIGRH, which means that discrepancies between SIGRH and the external real-time sources might not be detected. Implementing the proposed framework would strengthen CA by incorporating textual evidence, such as those from DOE.

4.2. Official gazettes[8]

DOEs are newspapers created, maintained, and administered by governments to publish official acts of the Executive branch, Legislative branch, and Judiciary branch (Wikipedia, 2023). In Brazil, the vast majority of DOEs are available online. Essentially, anything that has a public interest should be published in the DOE. Typically, the DOE is published daily in a portable document format (PDF), available online and accessible to everyone. Although there are guidelines for structuring these documents, they are not standardized. Examples of information that must be publicly accessible in DOE include the appointment of public servants (NOMEAR),[9] the dismissal of public servants (EXONERAR), and punishments applied to public servants.

According to the internal auditors from the GAPES, DOE serves as the most relevant personnel appointment information source that is accessible to them. Despite GAPES's long-standing use of data-driven CA over SIGRH data, the review of the DOE remained a manual, non-continuous activity. As auditors emphasized, a daily sweep of DOE entries would be ideal to surface personnel acts in near real-time. However, the sheer volume of editions and limited staff time meant the team could only perform periodic, manual checks. This

[6] The small group test allows a loop back to the design process for refinement, aligning with the DSRM process (Peffers et al., 2007).

[7] Value converted using an exchange rate of R$ 4.95 to one US dollar.

[8] Diário Oficial in Portuguese.

[9] For instance, if someone wants to apply for a position as an Accountant in the Santa Catarina Government, they must first pass a civil service exam. In specific situations, certain servants may be appointed by nomination, such as State Secretaries. In both cases, the servant cannot effectively become part of the state workforce without first being officially appointed and listed in the DOE.

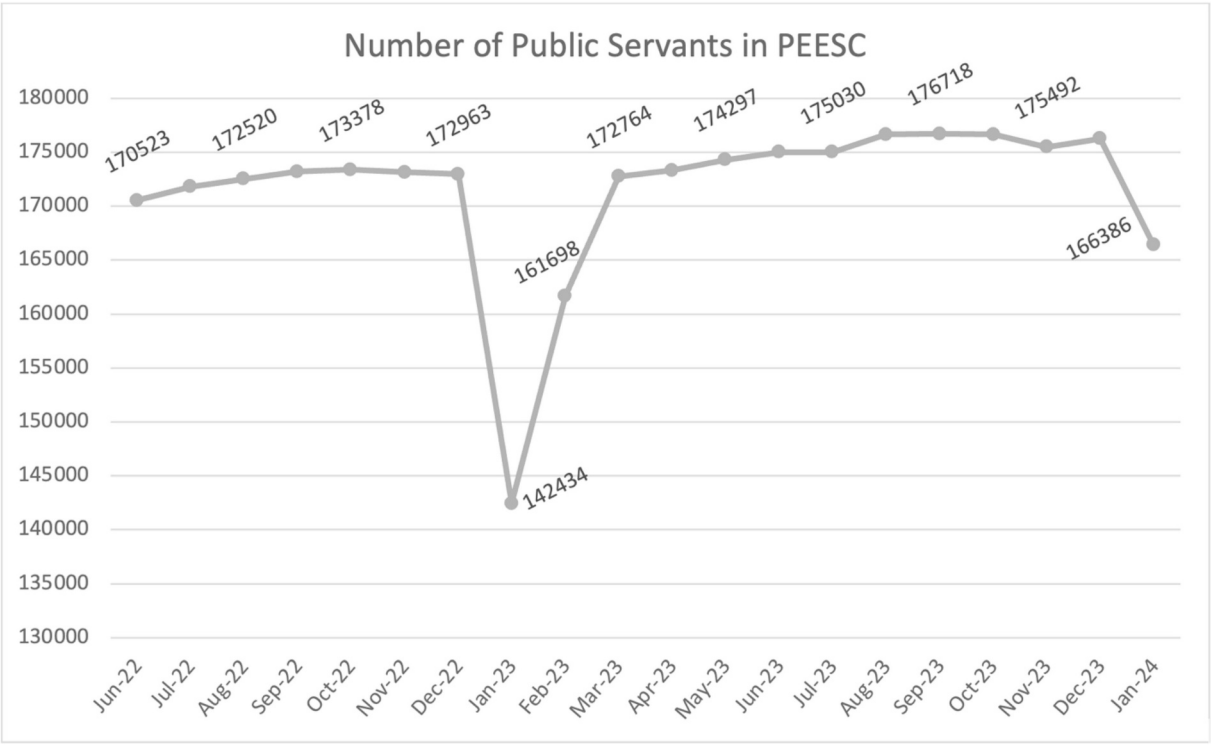


Fig. 4. Number of Public Servants in PEESC.

oversight might bring substantial risk to the payroll system. For instance, in its 2024 audit cycle, the Brazilian Federal Court of Accounts monitored the payrolls of 852 federal government organizations and identified 89,570 indicators of potential irregularities, which represented a monthly financial impact of approximately R$ 8.43 million (Tribunal de Contas da União, 2024). Furthermore, only 26,070 of the identified inconsistencies were clarified by the audited institutions during the period. These figures highlight the critical role of real-time auditing in preventing incorrect payments in payroll. The internal auditors emphasized that payroll, as the State's main expense, is highly material yet error-prone. Data are largely entered manually, the governing legislation is complex and voluminous, and the near-infinite variation in how events are described renders traditional rule-based methods ineffective. Fig. 4 shows the monthly variation in the total number of employees, reflecting the frequent events that affect payroll.

Therefore, cross-verification of payroll data with the DOE using the framework can complement the CA model already implemented at GAPES and generate savings for the State of Santa Catarina. In the following section, the proposed framework is illustrated, demonstrating its ability to extract and analyze audit evidence from the DOE and cross-verify the payroll information. The framework performs the following steps: 1) accesses the DOE website, downloads documents in PDF format, converts these documents to plain text, and conducts preprocessing; 2) defines the schema and extracts the audit evidence using LLM; 3) assists internal auditors in performing cross-verification with payroll data.

4.3. Application of the framework

The implementation of the framework involves the collaboration of internal auditors from the GAPES of the General Comptroller's Office of the State of Santa Catarina. The auditors assisted in identifying and validating audit evidence that could be extracted from the DOE and that could be useful as cross-verifying sources for payroll audits.

4.3.1. Acquiring, converting, and preprocessing (Step 1).

A robot is first programmed to automatically acquire the DOEs that are published daily by the state government website. This is done through an automatic scraping Python program. Fig. 5 shows the layout of the website that posts the DOE.[10] Specifically, the program leverages Selenium, an open-source software framework, to automatically analyze the infrastructure of the website and download the DOE (Debreceny et al., 2019). As the reports are posted daily, the program is set to be initiated accordingly for automatic downloading.

After acquiring, a data conversion step is performed to facilitate text preprocessing and context filtering, as the DOEs are encoded in

[10] The State of Santa Catarina's DOE website: https://portal.doe.sea.sc.gov.br/v2.40.00/#/portal/edicao/buscar-edicao

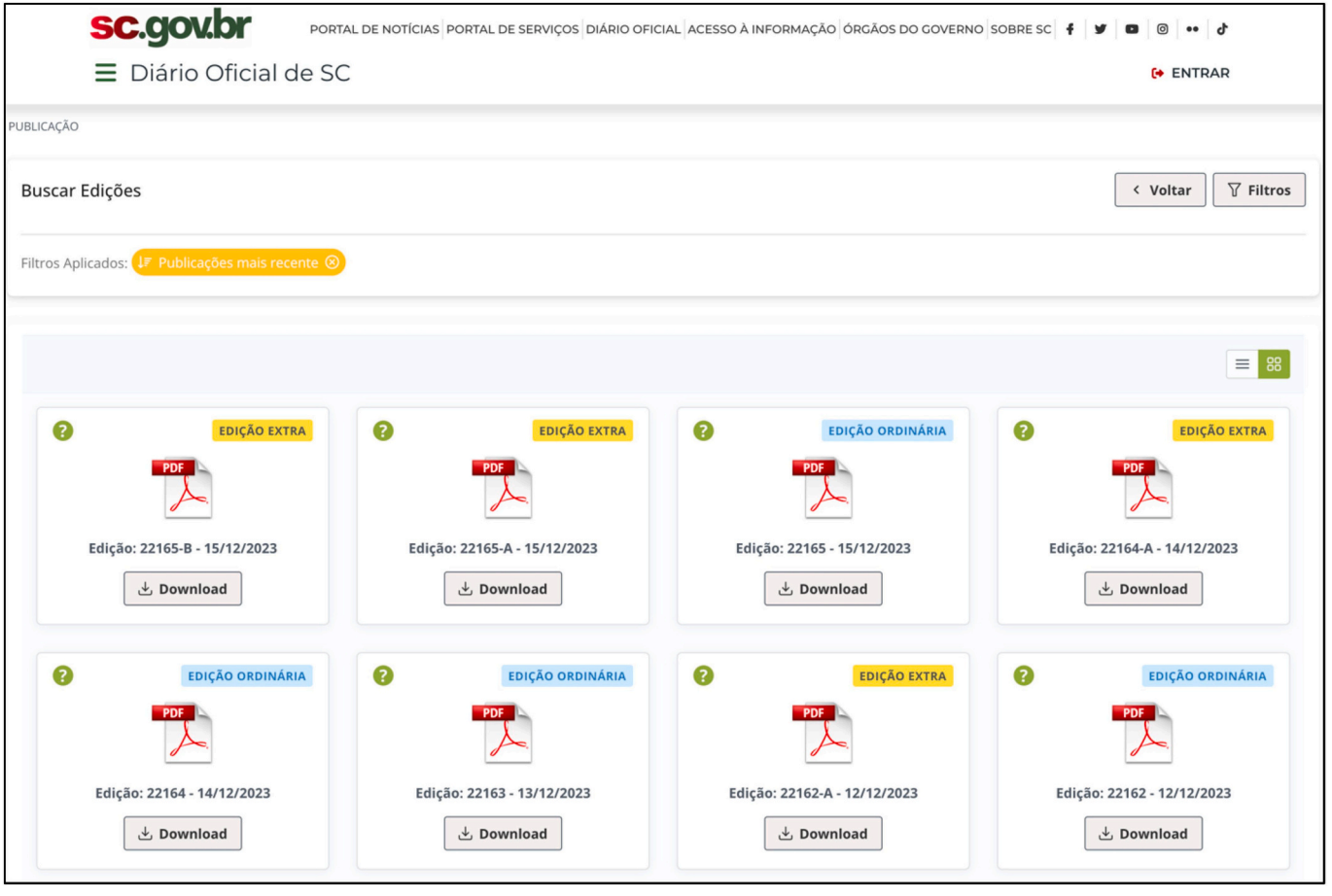


Fig. 5. Layout of the DOE Website.

PDF files. Specifically, this study supplies the acquired PDF files to "Cloudconvert" through an API to perform the PDF-to-TXT conversions.[11] Next, a set of text preprocessing functions is defined to standardize the data, including removing extra newlines and spaces and replacing certain Portuguese letters with their English counterparts to ensure output consistency.[12] The file conversion and preprocessing functions are embedded into the file-acquiring Python program to achieve seamless integration of the process, thus facilitating real-time analysis.

4.3.2. *LLM inference (Step 2)*

After preprocessing, the parsing schemas to obtain the audit evidence are defined ("Establishing parsing schema" in Fig. 2). For instance, employee appointment and dismissal details are valuable information for internal auditors to validate the payroll records. Appendix A shows two examples of the appointment and dismissal information from DOE. To parse these two types of employment information from the entire data file, this step first uses the keywords "NOMEAR" and "EXONERAR" to locate the relevant context.[13] Next, a prompt is constructed to identify the most relevant sentences from the isolated context ("Context filtering" in Fig. 2). The constructed prompt is provided in Appendix B.

This prompt instructs the LLMs to understand the given text based on its context, differentiate subtle nuances in the sentences, and provide reasonable interpretations. After this step, clean sentences relevant to appointments are isolated. Then, a schema of 18 variables was defined through collaboration between the co-author embedded with the audit team and the internal auditors from GAPES. The selection of these variables was driven by two criteria: (1) each variable must have a direct impact on payroll calculation or compliance verification, and (2) the variable must be extractable from DOE text with sufficient frequency to justify inclusion.[14] For example, errors or ambiguities in the registration numbers (*matrículas*) can generate incorrect payroll processing, while discrepancies in effective dates often lead to undue or retroactive payments. To ensure the accuracy of the schema, after defining the variables, the internal auditors reviewed them to confirm their relevance for cross-verifying the payroll system data.

Next, a structured instruction tuning with few-shot learning is used to design the prompts ("Prompt engineering" in Fig. 2). Appendix B summarizes the prompts for this demonstration. Following the principles of instruction tuning introduced in Section III, the prompts are designed by articulating explicit instructions in plain language.

The prompt design followed a structured development process informed by the audit prompt protocol proposed by Gu et al. (2024). Specifically, the prompt was organized as a chain of functionally distinct components: (1) a task explanation and role assignment that establishes the LLM's role as a structured information extractor for Portuguese administrative text; (2) explicit rules constraining the output format to JSON and prohibiting the addition of clarifying information; (3) input-output examples that operationalize few-shot learning, with eight examples selected to represent the major structural variations observed in DOE records (standard appointments, competitive exam appointments, military appointments, dismissals, and reversions); and (4) a task execution segment where the actual context is inserted for processing.

The development process was iterative. The initial prompt contained only the role assignment, rules, and two examples. Evaluation of early completions revealed three recurring failure modes, namely inconsistent output formatting (e.g., missing JSON fields), confusion between structurally similar fields (e.g., Internal Department versus State Department), and failure to extract information from less common record types (e.g., judicial-compliance appointments). In response, additional examples were introduced to cover these edge cases, and the rule section was strengthened to enforce strict JSON schema compliance. After four rounds of refinement, the prompt stabilized with eight examples covering the principal record variations, representative examples of which appear in Appendix B. The iterative approach aligns with best practices for prompt engineering in accounting applications (Gu et al., 2024; Li et al., 2025).

Regarding hyperparameters, the temperature was set to 0 to ensure deterministic outputs, which is critical for audit tasks where consistency and reproducibility are required. Other parameters were kept at their default values for the GPT-4 API.

With the prompts designed, the last step is to feed them with context into LLMs to perform the inference ("Prompt with context" in Fig. 2). GPT-4 from OpenAI is used as the LLM service provider to demonstrate this process. The API provided by OpenAI is used to establish a connection between the local CA system and GPT-4 ("Connect to LLM through API and perform the inference" in Fig. 2). This demonstration leverages the *API-RPA* LLM inference approach. By supplying the prompts through Excel files (emulating a local system), the RPA robot reads each of them and sends it to GPT-4 through API. GPT-4 will process the prompt with context and generate completion. Lastly, the RPA transmits the completions back to the internal system, such as the local Excel file in this demonstration.[15] Examples of the parsed audit evidence following the schema are shown in Fig. 6.

[11] See more information about Cloudconvert: https://cloudconvert.com/

[12] For instance, characters such as "À," "Á," "Ã," and "Â" are standardized into "A" to improve the consistency of textual outputs.

[13] Specifically, this step employs the Python Regex "{0,200}(NOMEAR|EXONERAR).{0,600}" to search the entire document. The results from the Regex ensure that 200 characters before the keyword ("NOMEAR" or "EXONERAR") and 600 characters after the keyword are retained as the context. To determine the context window containing the entire employment-related description, the authors manually test 20 documents and find out that the context window of 200 characters before the keyword ("NOMEAR" or "EXONERAR") and 600 characters after the keyword usually contains the entire employment-related information.

[14] These variables include (Employee) Name, Position, Function, Department, State/City, Matriculation ID (mtcl), Effective Date, Internal Document Number, Other Position Information, Branch, Legal Number, Internal Number, Workload, Person ID, Other Person ID, Position Type, Retirement Date, Retirement Publish Date, and (Dismissal) Reason.

[15] By leveraging RPA, the auditors can also input the parsed audit evidence into the government's internal ERP system for further verification or documentation purposes. For more details, refer to Huang and Vasarhelyi (2019).

FileName	Date	Name	Position	Function	Department	StateCity	mtcl	EffectiveDate
22207403082023	03 de agosto de 2023	MATEUS PEREIRA SILVA	Maj BM	chefe de Divisao de Investiga	DINVI	Florianopolis – SC	965876-9	12/07/2023
22207403082023	03 de agosto de 2023	GABRIELA RODRIGUES COSTA	3º Sgt BM	Sargenteante	3ª Companhia do 11º Batalhao de Bombeiro Militar (3ª/11	Capinzal – SC	954655-4	17/07/2023
22207403082023	03 de agosto de 2023	FERNANDO ALMEIDA SANTOS	TC BM	Comandante	Batalhao de Comando e Servicos do Quartel do Comando-(	Florianopolis – SC	985829-8	17/07/2023
22207403082023	03 de agosto de 2023	LUCIANA MORAES LOPES	3º Sgt BM	Sargenteante	1ª Companhia do 9° Batalhao de Bombeiro Militar (1ª/9º E	Canoinhas – SC	955544-1	17/07/2023
2199813042023	13 de abril de 2023	RODRIGO SOARES MACHADO	1º Ten BM	Comandante	2º Pelotao da 1ª Companhia do 1° Batalhao de Bombeiro M	Florianopolis – SC	922118-4	03 de abril de 2023
2199813042023	13 de abril de 2023	JULIANA VASCONCELOS LIMA	Cap BM	Comandante	3ª Companhia do 1° Batalhao de Bombeiro Militar (3ª/1º E	Florianopolis – SC	933802-5	03 de abril de 2023
2199813042023	13 de abril de 2023	ANDRÉ FONSECA MARTINS	Cap BM	Chefe Interina	7ª Secao do Estado-Maior Geral	Florianopolis – SC	900230-9	31/01/2023
2199813042023	13 de abril de 2023	CARLA DIAS BARBOSA	TC BM	Chefe	7ª Secao do Estado-Maior Geral	Florianopolis – SC	975546-5	31/01/2023
2199712042023	12 de abril de 2023	BRUNO OLIVEIRA FREITAS	Cap BM	Comandante	1º, 2º e 3º Pelotoes da 2ª Companhia do Centro de Formac	Florianopolis – SC	951070-6	27/02/2023
2195607022023	07 de fevereiro de 2023	SOFIA CARVALHO GONÇALVES	TC BM	Comandante Interino	1ª Regiao Bombeiro Militar (1ª RBM)	Balneario Camboriu – SC	923969-8	31 de janeiro de 2023
2195607022023	07 de fevereiro de 2023	MATEUS PEREIRA SILVA	Cel BM	Ajudante-Geral, Chefe de Gal	Comando-Geral do Corpo de Bombeiros Militar de Santa C	Florianopolis – SC	936599-4	31/01/2023
2195607022023	07 de fevereiro de 2023	GABRIELA RODRIGUES COSTA	1º Sgt BM	Comandante	1º Grupo do 1º Pelotao da 5ª Companhia do 7° Batalhao d	Ilhota – SC	986827-1	03/02/2023
2195607022023	07 de fevereiro de 2023	FERNANDO ALMEIDA SANTOS	TC BM	Comandante	15º Batalhao Bombeiro Militar (15° BBM)	Rio do Sul – SC	951420-8	13/02/2023
2195607022023	07 de fevereiro de 2023	LUCIANA MORAES LOPES	TC BM	Comandante	8° Batalhao de Bombeiro Militar (8º BBM)	Tubarao – SC	956247-4	22/02/2023
2195607022023	07 de fevereiro de 2023	RODRIGO SOARES MACHADO	TC BM	Comandante	8° Batalhao de Bombeiro Militar (8º BBM)	Tubarao – SC	933679-1	22/02/2023
2195607022023	07 de fevereiro de 2023	JULIANA VASCONCELOS LIMA	Maj BM	Comandante	2ª Companhia do 9° Batalhao de Bombeiro Militar (2ª/9° B	Sao Bento do Sul – SC	971600-3	13/02/2023
2195607022023	07 de fevereiro de 2023	ANDRÉ FONSECA MARTINS	Maj BM	Comandante	4ª Companhia do 9° Batalhao de Bombeiro Militar (4ª/9º E	Mafra – SC	970102-4	08 de marco de 2023
2195607022023	07 de fevereiro de 2023	CARLA DIAS BARBOSA	Cap BM	Comandante	1ª Companhia do 10º Batalhao de Bombeiro Militar (1ª/10	Sao Jose – SC	938465-9	12/12/2022
2195607022023	07 de fevereiro de 2023	BRUNO OLIVEIRA FREITAS	Cap BM	Comandante	1ª Companhia do 10º Batalhao de Bombeiro Militar (1ª/10	Sao Jose – SC	945866-5	16/01/2023
22206826072023	26 de julho de 2023	SOFIA CARVALHO GONÇALVES	ST BM	Sargenteante	3ª Companhia do 8º Batalhao de Bombeiro Militar (3ª/8º E	Braco do Norte – SC	953330-7	17/07/2023
22206826072023	26 de julho de 2023	MATEUS PEREIRA SILVA	2º Sgt BM	Sargenteante	3ª Companhia do 8º Batalhao de Bombeiro Militar (3ª/8º E	Braco do Norte – SC	914038-7	17/07/2023
2211228092023	28 de setembro de 2023	GABRIELA RODRIGUES COSTA	Cb BM	Comandante	4º Grupo do 3º Pelotao da 2ª Companhia do 2° Batalhao d	Salto Veloso – SC	960759-4	28/09/2023
2211228092023	28 de setembro de 2023	FERNANDO ALMEIDA SANTOS	Maj BM	Comandante	1ª Companhia do 12º Batalhao de Bombeiros Militar (1ª/1	Sao Miguel do Oeste – SC	978161-9	11/08/2023
2211228092023	28 de setembro de 2023	LUCIANA MORAES LOPES	3º Sgt BM	Comandante	2º Grupo do 2º Pelotao da 3ª Companhia do 7° Batalhao d	Balneario Barra do Sul – SC	933919-2	24/07/2023
2211228092023	28 de setembro de 2023	RODRIGO SOARES MACHADO	3º Sgt BM	Comandante	2º Grupo do 2º Pelotao da 3ª Companhia do 7° Batalhao d	Balneario Barra do Sul – SC	916950-6	24/07/2023
22208924082023	24 de agosto de 2023	JULIANA VASCONCELOS LIMA	Cb BM	Comandante	1° Grupo do 2° Pelotao da 1ª Companhia do 11° Batalhao (	Catanduvas – SC	996228-1	01/08/2023
2207910082023	10 de agosto de 2023	ANDRÉ FONSECA MARTINS	1º Sgt BM	Comandante	1° Grupo do 1° Pelotao da 1ª Companhia do 8° Batalhao d	Capivari de Baixo – SC	999057-6	16/06/2023
2207910082023	10 de agosto de 2023	CARLA DIAS BARBOSA	Maj BM	Comandante	1ª Companhia do 6° Batalhao de Bombeiro Militar (1ª/6º E	Chapeco – SC	975657-2	01/08/2023
2207910082023	10 de agosto de 2023	BRUNO OLIVEIRA FREITAS	TC BM	Comandante	Batalhao de Ajuda Humanitaria	Florianopolis – SC	949719-3	01/08/2023
2207910082023	10 de agosto de 2023	SOFIA CARVALHO GONÇALVES	Maj BM	Comandante Interino	Batalhao de Ajuda Humanitaria	Florianopolis – SC	920307-3	17/07/2023
2194319012023	19 de janeiro de 2023	MATEUS PEREIRA SILVA	Major BM	Comandante	2ª Companhia do Centro de Ensino Bombeiro Militar (CFAI	Florianopolis-SC	963949-1	31/12/2022
2194319012023	19 de janeiro de 2023	GABRIELA RODRIGUES COSTA	Cap BM	Comandante	4ª Companhia do Centro de Ensino Bombeiro Militar (CFCT	Florianopolis-SC	931945-1	31/12/2022
2194319012023	19 de janeiro de 2023	FERNANDO ALMEIDA SANTOS	1º Ten BM	Comandante	2º Pelotao de Comando e Servicos do Comando-Geral do C	Florianopolis – SC	900147-1	30/12/2022
2194319012023	19 de janeiro de 2023	LUCIANA MORAES LOPES	1º Ten BM	Comandante	2º Pelotao de Comando e Servicos do Comando-Geral do C	Florianopolis – SC	907841-8	30/12/2022
2194319012023	19 de janeiro de 2023	RODRIGO SOARES MACHADO	Tenente-Coronel BM	Sub Chefe do Estado Maior G	Corpo de Bombeiros Militar de Santa Catarina	Florianopolis – SC	986968-5	1 de dezembro de 2022
2195809022023	09 de fevereiro de 2023	JULIANA VASCONCELOS LIMA	Cap BM	Comandante	1° Pelotao da 3ª Companhia do Batalhao de Operacoes Aer	Florianopolis – SC	995669-4	02 de janeiro de 2023
2195809022023	09 de fevereiro de 2023	ANDRÉ FONSECA MARTINS	2º Sgt BM	Sargenteante	2ª Companhia do 7° Batalhao Bombeiro Militar (2ª/7ª BBM	Navegantes – SC	916525-9	27 de janeiro de 2023
2195809022023	09 de fevereiro de 2023	CARLA DIAS BARBOSA	Maj BM	Chefe Interino	Divisao de Logistica (DiL)	Florianopolis – SC	968751-1	31 de janeiro de 2023
2195506022023	06 de fevereiro de 2023	BRUNO OLIVEIRA FREITAS	TC BM	Comandante	7° Batalhao de Bombeiro Militar (7º BBM)	Itajai – SC	995455-6	27/01/2023
2195506022023	06 de fevereiro de 2023	SOFIA CARVALHO GONÇALVES	TC BM	Comandante	7° Batalhao de Bombeiro Militar (7º BBM)	Itajai – SC	912139-4	27 de janeiro de 2023
2195506022023	06 de fevereiro de 2023	MATEUS PEREIRA SILVA	TC BM	Comandante	13º Batalhao Bombeiro Militar (13° BBM)	Balneario Camboriu – SC	976356-2	27/01/2023
2195506022023	06 de fevereiro de 2023	GABRIELA RODRIGUES COSTA	TC BM	Diretor Interino de Logistica	Corpo de Bombeiros Militar de Santa Catarina	Florianopolis-SC	948702-9	31 de janeiro de 2023
2196623022023	23 de fevereiro de 2023	FERNANDO ALMEIDA SANTOS	ST BM	Comandante	1º Grupo do 2º Pelotao da 1ª Companhia do 6° Batalhao d	Chapeco – SC	985833-8	06 de fevereiro de 2023
2196623022023	23 de fevereiro de 2023	LUCIANA MORAES LOPES	Cb BM	Comandante	1° Grupo do 1° Pelotao da 1ª Companhia do 4° Batalhao de	Sideropolis – SC	966955-9	01 de fevereiro de 2023

Fig. 6. Examples of the Parsed Audit Evidence.

Table 1
Quantitative Evaluation Results.

Audit Evidence	Percentage of TP	Percentage of TN	Percentage of FP	Percentage of FN	Total Number of Evaluations
Name	100%	0%	0%	0%	150
Position	93%	7%	0%	0%	150
Function	27%	72%	0%	1%	150
Internal Department	52%	43%	5%	0%	150
State City	28%	70%	0%	2%	150
Mtcl	42%	53%	5%	0%	150
Effective Date	38%	62%	0%	0%	150
Internal Document Number	56%	37%	3%	4%	150
State Department (Branch)	61%	32%	7%	0%	150
Average	55%	42%	2%	1%	–

Note: This table presents the quantitative evaluation results of the framework. True positives (TP) are cases where the information is present in the DOE and is correctly parsed by the framework. True negatives (TN) are cases where the DOE does not contain that specific information, and the framework correctly identifies this, thus returning empty values. False positives (FP) are instances where the framework parses the wrong information from the DOE. False negatives (FN) occur when the DOE contains the desired information, but the framework fails to identify it adequately, returning an empty value. The percentage of each type of result is calculated by dividing the number of cases in that type by the total number of evaluations.

4.3.3. Cross-verification (Step 3)

Finally, in *Step 3*, the internal auditors retrieve the audit evidence parsed from the DOE and perform cross-verification with the internal payroll system data. This involves matching employee's name and identification number with the internal records to identify inconsistencies. For this demonstration, all internal payroll system data examined complies with the evidence parsed from the DOE. This provides an additional layer of assurance for the payroll data. Where an exception arises, the internal auditors perform manual checks to determine its underlying cause. The findings from the demonstration also serve as the basis for further refining the framework. For instance, the information about why an employee is dismissed, as disclosed in the DOE, is a valuable variable to be included in cross-verification. Consequently, the "(Dismissal) Reason" has been added to the schema as a revision to the framework following the initial illustration.

5. Evaluation of the results

5.1. Quality of the parsed audit evidence

To gauge the quality of the parsed audit evidence, the implementation results were manually examined against the original DOE text. The framework was applied to 131 DOE files spanning approximately 26 weeks of publications, yielding a total of 746 parsed records.[16] These files varied substantially in length and complexity. The average file was approximately 54 pages (median: 44), ranging from 2 to 280 pages, with an average of 5.69 extracted personnel records per file. This variation provides a natural distribution of document complexity, as longer files contain more personnel actions interspersed with unrelated content, increasing the cognitive and computational demands of accurate extraction.

To evaluate quality, 150 cases were randomly selected from the 746 parsed records (approximately 20% of the total population) for manual verification. The evaluation was conducted in two rounds, an initial round of 100 cases, followed by an additional 50 cases added to strengthen the assessment. Accuracy remained consistent across both rounds, indicating stability in the framework's performance. Each case was evaluated by one co-author working alongside two internal auditors from the GAPES team. The original DOE text served as the gold standard against which parsed fields were compared. The co-author and the auditors agreed on the correctness of the parsed values in the large majority of cases on first inspection, with fewer than 5% requiring discussion to reach consensus (a raw, pre-discussion agreement of approximately 95% or higher). These disagreements did not concern whether a documented fact was true or false but reflected field-boundary judgments, such as whether a particular string should be assigned to one field rather than an adjacent one (for example, Function versus Position) or whether a partial match should count as correct. These cases were resolved through joint review and consensus among the co-author and the auditors. The evaluation focused on the fields most significant for internal audit cross-verification purposes.

The quantitative evaluation results are outlined in Table 1. True positives (TP) were identified as cases where the information was present in the DOE and was correctly parsed by the framework. True negatives (TN) were cases where the DOE did not contain that specific information, and the framework correctly identified this, thus returning empty values in its parsed variable. False positives (FP) were cases where the framework parsed the wrong information from DOE. False negatives (FN) occurred when the DOE contained the desired information, but the framework failed to identify it adequately, returning an empty value.

[16] It should be noted that the prompts were developed using a separate subset of early documents distinct from the 131 files used for testing. Specifically, approximately three early DOE issues published before this 26-week testing window were used for development; because the task targets appointment and dismissal notices, these earlier issues necessarily contain the same document types, but they served only to design and refine the prompts and schemas, not to assess accuracy, so that no test file informed prompt development.

Table 2
Comparison of Regex and LLM Extraction Accuracy.

Audit Evidence	Regex				LLM			
	TP%	TN%	FP%	FN%	TP%	TN%	FP%	FN%
Name	86%	0%	0%	14%	100%	0%	0%	0%
Position	90%	8%	0%	2%	93%	7%	0%	0%
Function	10%	76%	12%	2%	27%	72%	0%	1%
Internal Department	36%	44%	0%	20%	52%	43%	5%	0%
State City	22%	76%	0%	2%	28%	70%	0%	2%
Mtcl	26%	64%	0%	10%	42%	53%	5%	0%
Effective Date	30%	66%	4%	0%	38%	62%	0%	0%
Internal Document Number	18%	38%	0%	44%	56%	37%	3%	4%
State Department (Branch)	76%	16%	6%	2%	61%	32%	7%	0%
Average	44%	43%	2%	11%	55%	42%	2%	1%

Note: This table compares the extraction accuracy of the Regex baseline (50 test cases) and the LLM-based framework (150 cases, from Table 1) using the same evaluation criteria. True positives (TP) are cases where information is present and correctly parsed. True negatives (TN) are cases where information is absent and correctly returned empty. False positives (FP) are cases where incorrect information was parsed. False negatives (FN) are cases where present information was not properly parsed. The percentage of each type of result is calculated by dividing the number of cases in that type by the total number of evaluations.

The framework performed very well, with an overall accuracy rate (both TP and TN) of 97%.[17] To better understand the conditions under which the approach fails and the implications of those failures, a detailed error analysis of FP and FN across high-impact fields is conducted. A notable FN pattern emerged in the *Internal Document Number* field (4% FN). An analysis of these instances revealed that the standard format of the process is the name of the *State Department (Branch)* + the year + a number. In some cases, the framework failed to identify that a particular entry was an internal process when the name of the *State Department* was not included in DOE. While human auditors intuitively infer the department from the broader context, the LLM missed this detail, returning an empty value. For FP, one critical error type occurs in organizational units. The framework occasionally misclassified the *Internal Department* with the *State Department* (7% FP), an error that even a human auditor unfamiliar with the entire state structure could make. Another example arises from identifier ambiguity in administrative acts. In a judicial-compliance appointment record, the text includes "inscrição n° 127926," which denotes the applicant's registration number in the public competitive examination rather than the employee's payroll/HR registration number (Matrícula). Because the string resembles typical Matrícula formats (short numeric identifier) and appears alongside other administrative identifiers, the framework incorrectly populated the Matrícula field with the public examination registration number. FP might trigger unnecessary investigation or, worse, mask the absence of true information if reviewers assume the extracted value is authoritative. However, even trained auditors can be misled when multiple similar information occurs in the same excerpt. We hypothesize that fine-tuning the LLM on the existing state architecture documents or employee administrative database from Santa Catarina could reduce such errors. Testing this conjecture is left to future research.

It is also worth noting that this evaluation focuses on the extraction accuracy of *Step 2* (LLM inference), as the primary objective of the framework is to generate structured audit evidence that serves as input for the CA process. The cross-verification step (*Step 3*) employs deterministic rule-based matching, such as comparing parsed fields like employee names and identification numbers against payroll records, whose reliability depends directly on the quality of the extracted inputs. As demonstrated, the 97% extraction accuracy provides a strong foundation for downstream cross-verification. Nonetheless, extraction errors propagate differently depending on the field affected. Errors in the cross-verification keys, the employee name and Matrícula, are catastrophic. An incorrect key can silently produce a wrong or failed match, as in the Matrícula misclassification noted above (5% FP), although the name field showed no such errors (0%). Errors in descriptive, non-key fields are comparatively benign. A misclassified Internal Department or State Department field (5% and 7% FP, respectively) or an Internal Document Number error (3% FP, 4% FN) does not corrupt the primary match and instead surfaces as a reviewable exception or affects only secondary analyses. Assurance effort should therefore concentrate on the identifier fields that anchor the cross-verification. Future research could empirically quantify end-to-end false-alarm rates at scale and the operational cost of investigating false exceptions generated by the complete process.

The auditors' qualitative evaluation was broadly positive, characterizing the proposed framework as a crucial control that extends coverage and improves timeliness without imposing significant recurring costs or heavy staffing demands. They highlighted opportunities to strengthen the framework by applying natural language processing not only to DOE extracts but also to internal databases, thereby enabling robust record linkage in the presence of name and department variations and accommodating lags between DOE publication and system updates.

To assess whether the use of LLMs is justified relative to traditional text-mining methods, we conducted a baseline comparison using optimized Regex. Twenty randomly selected records were used for Regex pattern development, and a separate set of 50 records was used for testing. Comprehensive Regex rules were designed for each of the nine key fields with full visibility into the data formats. Table 2 reports the results. The Regex approach achieves an overall accuracy of 87%, compared to 97% for the LLM-based

[17] The 3% reflects a one-time accuracy benchmark from the 150-case verification; in production, Step 3's rule-based cross-verification automatically flags records conflicting with the payroll data as exceptions for review, so auditors need not re-read the full DOE.

framework. The gap is most pronounced in audit-critical fields. *Internal Document Number* accuracy is only 56% with Regex, primarily because minor punctuation variations (e.g., "n.°" versus "n°" versus "n°") and non-standard formatting of process numbers cause patterns to fail. *Name* extraction achieves 86% because the position of the name varies across record types, and no single Regex pattern set can accommodate all variations. *Department* extraction is 80%, as the Regex requires a predefined keyword list that misses new organizational units and nested references.

Notably, the *Function* field reveals a qualitative limitation of Regex that accuracy metrics alone may understate. While the LLM achieves 99% accuracy, the Regex achieves only 86%, with 12% of cases being FP, incorporating irrelevant content such as city names, workload details, or process numbers into the field. Such boundary errors would corrupt downstream cross-verification.

Regarding the LLM implementation cost, the estimated annual LLM API cost is approximately $237 based on GPT-4 pricing as of 2023.[18] For a payroll system disbursing over $500 million per month, this cost is negligible. It is also worth noting that while Regex-based approaches impose ongoing maintenance costs as new textual variations emerge, the LLM approach accommodates such variations through natural language instructions without requiring pattern updates.

5.2. Effectiveness of the framework

The implementation reveals that this framework acts as an important improvement in internal auditors' current CA model. Although GAPES has had a functioning CA process for over 10 years, this is the first time the auditors have used more advanced techniques (e.g., LLMs) for CA purposes. Before the implementation, the auditors could only manually perform the cross-verification using random sampling methods, which could cause significant oversight. The framework allows internal auditors to conduct the full population testing (Freiman et al., 2022), thus creating enhanced control for the State without incurring significant costs or the need for intensive human labor. This is extremely important since most entities usually have limited financial and human resources for these activities.

In addition to the benefits, a concern was raised regarding the implementation of the framework, namely the reliability of the information parsed using this framework (O'Leary, 2023a), as manual and continuous validation of the results is impractical. Specifically, situations might exist where the parsed evidence does not correspond to reality or is omitted from the analyses (FP or FN).[19]

The parsed audit evidence from the framework still lacks perfection, such as the existence of FP and FN. Although several prompt engineering approaches have been used to enhance inference consistency and accuracy, errors still occur. As this is still in the early stage of LLM development, it is foreseen that the advancement of the LLM could further enhance the accuracy of the results. Additionally, we hypothesize that a specialized LLM fine-tuned on the internal data could help alleviate this concern by generating more accurate results, a possibility that future research should test. With the help of RPA (Huang and Vasarhelyi, 2019; Zhang, 2023), an automatic process that is unattended by human auditors could address the limitation of lack of personnel from the internal audit team.

5.3. Efficiency of the framework

According to the internal auditors, the current method for cross-verifying payroll data using DOE is predominantly manual. Therefore, even straightforward tasks such as cross-verifying names and positions become time-consuming, given that each of the DOEs may contain hundreds of payroll-related publications.[20] To determine the time efficiency of the proposed framework in facilitating the cross-verification of accounting information, an evaluation was undertaken. This evaluation compared the average time required to manually analyze 100 randomly selected cases in the DOE against the time the framework takes to cross-verify an equivalent number of cases.[21]

Specifically, the process includes accessing the DOE website, acquiring and converting the document, searching for the relevant exogenous variables, accessing the Payroll Management System, and cross-verifying basic information such as the public servant's ID, the internal department, and the date of appointment or dismissal. On average, it took 3 min to complete these steps manually (the auditors' current primary approach) for each piece of information by one experienced auditor. Comparatively, the framework takes less than 30 s to complete the same cross-verification task.[22] This includes automatic acquisition, conversion, and preprocessing of a DOE document, extracting critical variables using LLM via an API, and finally accessing the local ERP system (Payroll Management System) through RPA to perform rule-based examinations. This represents an approximately 83% reduction in time cost. Continuous implementation of this framework could lead to profound and lasting enhancements in operational productivity, ensuring more effective CA and allocation of resources. It should be noted that this efficiency estimate does not account for the time auditors may spend

[18] The estimated annual API cost is approximately $237, based on GPT-4's 2023 pricing; under current model pricing this figure is likely conservative and should be read as an upper bound. This annual figure is illustrative: it extrapolates from an observed average of 5.69 records per DOE file and assumes a stable document volume of approximately 1500 records per year. The detailed token-level derivation is provided in Online Appendix B. This estimate covers only API usage and does not include infrastructure, maintenance, or human oversight costs.

[19] For example, an act of appointment naming a specific public servant, but the framework for some reason generates completely different information or misses that piece of record.

[20] This process requires auditors to manually navigate through the DOE web pages, open relevant files, and visually search for payroll-related information. Following this, they have to enter the system, input data concerning the specific public servant, and sift through numerous reports in search of any potential inconsistencies.

[21] The time associated with the development of the framework (e.g., establishing schema and prompt engineering) is not considered during the efficiency evaluation, as it represents a one-time fixed cost and could become trivial if the framework is operated continuously in the long run. Additionally, this development phase could be treated as part of the framework's learning process, similar to how a human auditor undergoes training before performing their audit tasks.

[22] Based on the processing efficiency of GPT-4 as of December 2023.

Table 3
Evaluation of the Proposed Framework.

Dimension	Measure	Result	Contribution
Effectiveness	Scope of testing	Full-population vs. sample-based testing	Better anomaly detection
Efficiency	Time to cross-verify one record	Reduced from 3 min to 30 s	Up to 83% reduction in processing time
Quality	Accuracy of parsed audit evidence	97% average accuracy	Ensures high reliability of extracted audit evidence

Note: This table summarizes the performance of the proposed framework in terms of effectiveness, efficiency, and quality.

investigating FP and FN generated by the framework. To the extent that such cases require manual follow-up, the net efficiency gain would be lower than 83%.

Additionally, the examples shown above are the simplest cross-verification tasks. More complex ones, such as assessing whether the remuneration is consistent with the function of the public servant, would take a longer time for auditors since it requires consulting various relevant legislation. In those cases, the framework would be more efficient as the LLM can parse critical evidence from a large amount of real-time text more efficiently than human auditors. All of the evaluation results are summarized in Table 3.

6. Discussion and conclusion

This study proposes an LLM-assisted framework that can identify audit evidence from real-time textual sources to cross-verify the accounting data, thus enhancing the effectiveness, efficiency, and quality of the CA application. The framework focuses on the implications for auditors as they adopt disruptive technologies to collect unstructured data and transform it into formats suitable for CA procedures. Beyond the demonstrated artifact, this study yields several generalizable insights. From a theoretical perspective, the framework can be understood through the lens of information-processing theory (Galbraith, 1974). Continuous auditing of textual data imposes substantial information-processing demands on auditors (e.g., searching, reading, interpreting, and reconciling large volumes of heterogeneous text against accounting records). The three-step framework reduces these demands by automating the acquisition and standardization of textual inputs (Step 1), offloading the cognitively intensive task of parsing granular evidence to LLMs guided by structured schemas (Step 2), and channeling auditor attention toward exception investigation rather than routine extraction (Step 3). In this view, the schema functions as an uncertainty-reduction mechanism that converts ambiguous textual signals into structured, comparable data points.

Drawing on the demonstration and evaluation, we propose three design principles for LLM-assisted CA frameworks. First, schema-guided extraction with auditor-defined fields is preferable in high-stakes, formalized settings where auditability, provenance, and integration with existing rule-based systems are required. Second, context filtering before LLM inference, such as using domain keywords to isolate relevant passages, improves both accuracy and cost efficiency, and should be adopted whenever the target information can be identified through predictable lexical markers. Third, rule-based cross-verification as a downstream validation layer ensures that LLM outputs are not accepted at face value but are systematically compared against authoritative records, preserving auditor judgment in the loop.

These principles are most effective under specific boundary conditions. The framework excels when textual evidence is high-volume, semi-structured, and governed by standardized legal or administrative conventions, as is the case with government gazettes, procurement announcements, and standardized contract filings. Its effectiveness may diminish with free-form correspondence, noisy optical character recognition outputs, or multi-jurisdictional legal texts where terminology and conventions vary substantially, requiring more adaptive preprocessing and potentially different prompting strategies.

In practice, this points to specific, well-suited applications such as contract compliance reviews, procurement cross-verification, and loan agreement validation, where standardized filings can be checked in real time against internal transactions to prevent overpayments or fraud. By clearly defining these boundary conditions, auditors can leverage the framework where it adds the most value without underestimating the complexities of less structured textual sources.

As business activities are becoming increasingly digitized, obtaining real-time audit evidence from textual data becomes essential for CA applications. Through a three-step LLM-assisted framework, this study introduces the process of parsing critical audit evidence from the text in real time to cross-verify accounting data. The demonstration shows that the framework can achieve the desired effectiveness, maintaining an average accuracy of 97% for the parsed audit evidence, and a reduction of up to 83% in cross-verification time.

This study contributes to accounting literature by proposing a framework that can obtain real-time audit evidence from textual data, overcoming the limitations of existing methods in parsing granular audit evidence from text. It also contributes to the growing literature on applying technology to enhance continuous audit. We operationalize an LLM inside our continuous auditing framework to continuously transform heterogeneous textual sources into structured data. This step is essential for scale. Without it, auditors cannot feasibly perform full population testing on text-based evidence. As fraud is rarely revealed by random sampling, especially when the relevant evidence is unstructured text, integrating LLMs with a continuous model is therefore a practical necessity for effective fraud analytics on textual data. Furthermore, the value of this study lies not only in demonstrating the capabilities of LLMs but also in structuring their application into a replicable CA framework that auditors can realistically adopt, ensuring that technological efficiency yields effectiveness, replicability, and audit quality beyond the simple application of LLMs.

However, this research has several limitations. First, the demonstration is conducted within a single governmental payroll setting. Further research is needed to validate the framework in various scenarios and audit tasks as suggested above. Additionally, the parsed

audit evidence lacks perfection. Although the average accuracy reaches 97%, we hypothesize that fine-tuning the framework on documents relevant to the audit task could further improve accuracy, a possibility that future work should test. Future iterations of the parsing schema can also incorporate confidence scores and evidence citations specifically for fields with relatively low accuracy and high audit impact to enhance the trustworthiness and explainability of the audit evidence. For long-term deployment, future research could develop prompt governance practices, such as version control and benchmark testing after each model update, since provider model updates can silently degrade extraction accuracy. Moreover, as LLMs possess multilingual capabilities, future research should provide direct evidence of cross-language extraction quality in cross-border audit settings. Finally, our field engagement at GAPES indicated that realizing the framework's benefits also depends on auditors' competencies, including data and analytical skills, a systemic understanding of the regulatory environment, and the ability to communicate results to management. A dedicated examination of these human-capital requirements for LLM-assisted CA is left to future research.

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Appendix A. Examples of the appointment and dismissal information from the DOE

NOMEAR o senhor FELIPE MIRANDA, portador do RG nº 8593421, inscrito no CPF nº 123.456.789-00, para exercer o cargo em comissao de Assessor de Comunicacao da Defensoria Publica do Estado de Santa Catarina, com efeitos a partir do dia 21/02/2022.

EXONERAR, conforme Art. 169 da Lei 6.745/85, a pedido, LUCAS SOUZA DE OLIVEIRA, matrícula 0981234-5-67, ocupante do cargo de Técnico Universitário de Execução da UDESC/CEFID, a contar de 20/10/2023, conforme processo UDESC 98765/2023.

Translated into English, the two pieces of employment-related information are as follows:

APPOINT Mr. FELIPE MIRANDA, bearer of ID card no. 8593421, enrolled under CPF no. 123.456.789-00, to assume the position of Communication Advisor for the Public Defender's Office of the State of Santa Catarina, with effect from February 21, 2022.

DISMISS, as per Article 169 of Law 6,745/85, at their request, LUCAS SOUZA DE OLIVEIRA, registration number 0981234-5-67, holding the position of University Technician at UDESC/CEFID, effective from October 20, 2023, according to UDESC process 98765/2023.

Note: The names, identification numbers (RG and CPF), and process/matriculation numbers in these examples have been changed to fictional equivalents to preserve the privacy of the actual individuals involved.

Appendix B. Prompts for the demonstration

You are an assistant skilled at extracting employment-related sentences from the CONTEXT in Portuguese. Based on the given CONTEXT, identify the full sentences that describe the appointment or dismissal of an employee. An example is: NOMEAR o senhor MARCOS PEREIRA, portador do RG nº 7623458, inscrito no CPF nº 987.654.321-00, para exercer o cargo em comissao de Assessor de Comunicacao da Defensoria Publica do Estado de Santa Catarina, com efeitos a partir do dia 21/02/2022.

[Role and Background] You are an assistant who is good at extracting structured information from the given CONTEXT in Portuguese.

[Rule] Do NOT add any clarifying information. The output MUST follow the JSON schemas shown in the examples.

[Tasks and Examples]

Example 1:

CONTEXT: NOMEAR o senhor MARCOS PEREIRA, portador do RG nº 7623458, inscrito no CPF nº 987.654.321-00, para exercer o cargo em comissao de Assessor de Comunicacao da Defensoria Publica do Estado de Santa Catarina, com efeitos a partir do dia 21/02/2022

OUTPUT: {
'Name': 'MARCOS PEREIRA',
'Position': 'Assessor de Comunicacao',
'Function': 'None',
'Department': 'None',
'StateCity': 'None',
'mtcl': 'None',
'EffectiveDate': '21/02/2022',
'InternalDoc': 'None',
'OtherPositionInfo': 'None',
'Branch': 'Defensoria Publica do Estado de Santa Catarina',
'LegalNo': 'None',
'InternalNo': 'None',
'WorkLoad': 'None',

'PersonID': 'RG nº 7623458',
'OtherPersonID': 'CPF nº 987.654.321-00',
'PositionType': 'em comissao',
'RetireDate': 'None',
'RetirePublishDate': 'None',
'Reason': 'None'
}
Example 2:
CONTEXT: NOMEAR POR CONCURSO, em cumprimento a decisao judicial proferida nos autos nº 5002055-66.2021.8.24.0159 e nos termos do processo FCEE 867/2022, MONICA SEIXAS DE OLIVEIRA MELLO, inscricao nº 2763, para exercer o cargo de provimento efetivo de PROFESSOR, nivel III, referencia A, do Quadro do Magisterio da Fundacao Catarinense de Educacao Especial - FCEE, com carga horaria de 40 horas, e exercicio em Instituicao Conveniada com a FCEE, na 36ª Regiao, nos termos do Edital FCEE nº 001/2014
OUTPUT: {
'Name': 'MONICA SEIXAS DE OLIVEIRA MELLO',
'Position': 'PROFESSOR',
'Function': 'None',
'Department': 'None',
'StateCity': 'None',
'mtcl': 'None',
'EffectiveDate': 'None',
'InternalDoc': 'None',
'OtherPositionInfo': 'nivel III, referencia A',
'Branch': 'Fundacao Catarinense de Educacao Especial – FCEE',
'LegalNo': 'nº 5002055-66.2021.8.24.0159',
'InternalNo': 'FCEE 867/2022',
'WorkLoad': '40 horas',
'PersonID': 'None',
'OtherPersonID': 'None',
'PositionType': 'None',
'RetireDate': 'None',
'RetirePublishDate': 'None',
'Reason': 'None'
}
Example 3:
CONTEXT: NOMEAR, para exercer a funcao de Comandante do 1º Grupo do 1º Pelotao da 2ª Companhia do 8° Batalhao de Bombeiro Militar (1º/1°/2ª/8º BBM), com sede em Imarui – SC, DIEGO SOUZA, 2º Sgt BM mtcl 922043-9, com efeitos a contar de 01 de janeiro de 2022
OUTPUT: {
'Name': 'DIEGO SOUZA',
'Position': '2º Sgt BM',
'Function': 'Comandante',
'Department': '1º Grupo do 1º Pelotao da 2ª Companhia do 8° Batalhao de Bombeiro Militar (1º/1°/2ª/8º BBM)',
'StateCity': 'Imarui – SC',
'mtcl': 'mtcl 922043-9',
'EffectiveDate': '01 de janeiro de 2022',
'InternalDoc': 'None',
'OtherPositionInfo': 'None',
'Branch': 'BM',
'LegalNo': 'None',
'InternalNo': 'None',
'WorkLoad': 'None',
'PersonID': 'None',
'OtherPersonID': 'None',
'PositionType': 'None',
'RetireDate': 'None',
'RetirePublishDate': 'None',
'Reason': 'None'
}
CONTEXT
[Insert the CONTEXT here]
Now start parsing the information.

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.accinf.2026.100786.

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